Mobicom Telematics JT808 Protocol V2.2

Applicable Models

Wiring product series: G102/G105/G106/G108/G109

Battery product series: G107/W100/W500/W600

GPS Lock series: GL500C/GL600C/GL800C

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Version History

Version	Update Date	What's new
V1.0	2019-10-07	Organize and publish the new protocol framework
V1.4	2023-6-19	Adjust the document structure, add 0x0300 command description and add examples
V1.41	2023-6-20	Add extra message definition 0xFC of LTE module's imei
V1.42	2023-6-24	Add extra message definition 0xFD of wiring tracker's built-in battery voltage
V1.43	2023-6-26	Alarm Indicator Define [Table 5-6] Update: After the event occurs, the alarm will only be reported once The main power supply of the terminal is powered off Acc on event Acc off event Status indicator definition [Table 5-7] Update: The main power supply of the terminal is powered off
V1.5	2023-6-29	Add The second configuration parameter method 0x8103 Add extra message definition 0x40 for ibutton & RFID function
V1.51	2023-07-07	Corrected and optimized the parsing of 0x0200
V1.6	2024-06-23	Add extension bytes 0xE7, 0xE8, 0xE9 0xE7: 24 types alarm status,0xE7 packet alarm status bit definition 0xE8: 24 types status,0xE8 packet switch status bit definition 0xE9: 1-100 battery percentage.1 byte. The same as 0xEE
V1.7	2024-08-01	Add extension bytes 0x61 0x61: External voltage of battery products
V1.8	2024-09-16	Add extension bytes 0x58 0x58: Humidity
V1.9	2024-10-18	Add extension bytes 0xD8,0xD9 for GPS Lock
V2.0	2025-1-17	Add extension bytes G109-MK 0xF4
V2.1	2025-7-11	Add Leave the fortified area alarm and status

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1 Preface

This file is to describe communication protocol between GPS padlock and platform. GPS padlock/terminal sent data via GPRS with using TCP. GPS padlock/terminal is the client, platform is the server.

2 Communication Connect

2.1 Build Connection

Terminal communicates with platform via GPRS by using TCP protocol. Terminal should connect to platform as soon as possible once power on. Terminal sends location & lock status info to platform immediately once connection set up.

2. 2 Keep Connection

After connection set-up, terminal should report its location/status to platform periodically, platform received and response to terminal, report interval time decided by terminal.

Note:

Keep connection is necessary if requires terminal ready to receive commands from platform at any time. If set interval report time is more than 120 seconds, and device does not report location/status info to platform after 120s, then need to send heartbeat packet, so that GPRS link will not be cut off as empty link by GSM operators.

2.3 Disconnect

Both platform and terminal can cut off connection based on TCP protocol, and both should initiatively

judge whether the connection cut off or not.

Ways for Platform to judge disconnection:

——According to TCP protocol, see if connection cut off by terminal;

——Terminal re-connect to platform with same ID, means the former connection be cut off;

——No message from terminal for some time, like heartbeat message.

Ways for Platform to judge disconnection:

——According to TCP protocol, see if connection is cut off by platform;

——GSM network linkage disconnected;

——GSM network normal, but no reply from platform after several times reporting.

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3 Message Processing

3. 1 Message From Platform

All messages sent from platform require terminal response. When the sender waits for a response timeout, the message will be re-sent. The response timeout and the re-send times are specified by the platform. Each re-send response timeout and

re-send times are specified by the platform. Below is the formula of the response timeout after each re-send:

$$T_{N+1}=T_N*(N+1) \qquad \text{Formula (1)}$$

Remark：

T_{N+1}: Response timeout after each re-send; T_N: Last response timeout;

N: Resend times

3.2 Message Sent From Terminal

3.2.1 Communication Link Connected

If communication link is connected, all messages sent from terminal require platform response. When the terminal waits for a response timeout, the message will be re-sent. The response timeout and the re-send times are specified by the terminal. Formula (1) can be used to calculate each resend response timeout after each re-send. If the terminal does not get any response from platform after reaching the setting times, then it will save the message locally.

3.2.2 Communication Link Disconnected

If communication link disconnected, terminal should save all message (call it blind-zone/history data) locally and resend to platform when it is connected again. For priority of the data, real-time data sent firstly, history (Blind-zone) data secondly.

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4 Protocol Basic

4.1 Communication Mode

It takes TCP/UDP platform as server end, and the terminal as client end. Normally, we use TCP protocol as main communication mode.

4.2 Data Type

Data type in protocol as below Table 4-1:

DATA TYPE	DESCRIPTION AND REQUIREMENT
BYTE	Unsigned single-byte integer (byte: 8-bit)
WORD	Unsigned double-bytes integer (byte: 16-bit)
DWORD	Unsigned four-bytes integer (double bytes: 32-bit)
BYTE[n]	N bytes
BCD[n]	8421 code， n byte

Table 4-1

4.3 Transmission Data Rule

The device transmits the **BYTE** and **WORD** data to server side by using Big-endian mode.

Rule as follow:

Transfer agreement of **BYTE**: transferred by byte stream.

Transfer agreement of **WORD**: transferred high 8 bits first step, and then low 8 bits.

Transfer agreement of **DWORD**: first high 24 bits, next high 16 bits, and then high 8 bits, and low 8 bits last.

4.4 Message Structure

Message Structure

Every piece data consists of packet header/end, message header, message body, and check code, the structure as below table 4-2:

packet header	Message header	Message body	Check code	packet end

Table 4-2

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4.5 Message Header

Content of message header as below **Table 4-3.**

START BYTE	WORD	DATA TYPE	INSTRUCTION
0	Message ID	WORD	
2	**Message body 's attribute**	WORD	The structure of Message body attribute. refer to **Table 4-4**
4	Terminal ID	BCD [6]	Defined by User
10	Message's serial number	WORD	Accumulated from zero by sending sequence

Table 4-3

Message body 's attribute structure .as below **Table 4-4**:

From 15th bit to 0th bit. From high bit to low bit.

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	Connection Property	Reserved				Length of message body									

Table 4-4

Connection Property Explain:

bit14: 0 mean long connection; 1 mean short connection;

Long Connection:

Long connection means device should connect to platform for all time, so that device can receive commands from platform in any time. When interval report time is more than 120 seconds, then need to send heartbeat packet to stop GSM network be disconnected as an empty connection.

Short Connection:

Short connection means device does not need to connect with platform in all time. It just needs to report to the platform in scheduled time or report alert. In that condition, platform cannot send commands to device in any time. Short connection can save battery energy, a battery can be keep used for several months or even one or two years.

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4.6 Packet Header/End And Escape

Use 0x7e， if there is 0x7e in check code, message header and message body, it will need to escape by following definition:

0x7e ←→0x7d 0x02；

0x7d ←→0x7d 0x01

Steps:

When Device send message: Message packing → count and fill check code → escaping

When server receive message: Restore escape character convert to real data → verify check code → analysis message

e.g.:

Send a message with content: 0x30 0x7e 0x08 0x7d 0x55, after packing, it will change to 0x7e 0x30 0x7d 0x02 0x08 0x7d 0x01 0x55 0x7e

4.7 Check Code

Check code occupy one byte. From **Message header** to **Message body**, XOR each bytes, and get check code.

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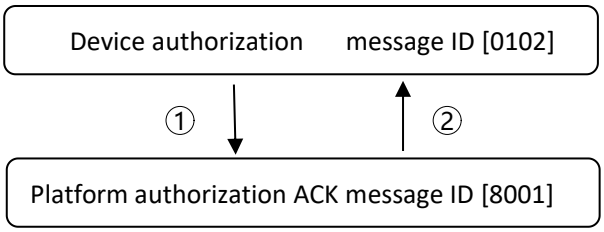
5 Uploading Data Format From Device & Server

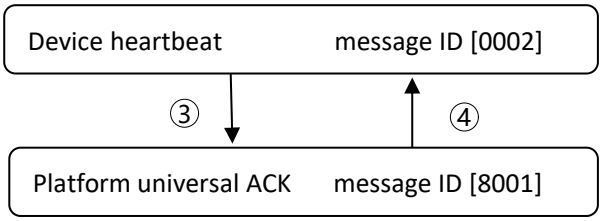
5.1 Communication Between Device And Platform

Device authentication： Device send message to platform, to verify itself.

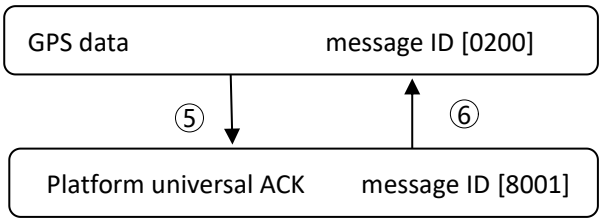
First step： Device authorization【0102】



Second step： Device heartbeat package【0002】



Third step： Device uploads GPS data【0200】



Note：

1) Only after the ACK of device authorization is successful, device will send heartbeat package data or GPS data to server side.

2) In second step, there's no order of priority about device heartbeat data and GPS data. Platform (server program) unify to use platform universal ACK **0x8001**

3) If under the same platform (reconfigure the device server IP and port), device restart or connection with platform break off, Device will request device authorization again.

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5.2 Platform Universal ACK【8001】

Message ID: 0x8001

For format of the platform universal ACK's message body content. Refer to **Table 5-1**

START BYTE	FIELD	DATA TYPE	DESCRIPTION
0	ACK serial number	WORD	Related device message serial number
2	ACK ID	WORD	Related the data's Message ID
4	result	BYTE	0: success/confirm 1: failed 2: message error 3: not support 4: alarm confirmation

Table 5-1

5.3 Device Authorization【0102】

Terminal uploads message ID: 0x0102

5.3.1 Format Of Device Authorization Message

START byte	Field	Data type	Description
0	Authorization code	STRING	terminal reconnect to upload authorization code

Table 5-2

Device Authorization data example:

7E01020004089022849248029F4A4C5854747E

Index	Value(HEX)	Instruction
1	**7E**	**packet header**
2	**0102**	**Message ID**
3	**0004**	**Message body 's attribute**
4	**089022849248**	**Terminal ID**
5	**029F**	**Message's serial number**
6	**4A4C5854**	**Authorization code (No need to process)**
7	**74**	**check code**
8	**7E**	**packet end**

Table 5-3

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5.3.2 Device Authorization Request- Platform Universal ACK example

Device authorization data example

7E01020004089022849248029F4A4C5854747E

e.g.: **4A4C5854 is Message body, the data length is 0004 bytes**

Platform Universal ACK data example

7E80010005089022849248 02A0029F0102005C7E

e.g.: **029F010200 is Message body, the data length is 0005 bytes**

Index	Value(HEX)	Instruction
1	7E	packet header
2	8001	Message ID
3	0005	Message body 's attribute. data content length
4	089022849248	Terminal ID
5	02A0	Message serial number
6	029F	Related Device authorization data's serial number. This serial number must be correct.
7	0102	Response data ID
8	00	Result： successfully
9	5C	Check code
10	7E	packet end

Table 5-4

5.4 Device Heartbeat Packet【0002】

Message ID: 0x0002

Server program need platform response【8001】to respond it

Terminal sending heartbeat data example

7E00020000008902284924802486 87E

Platform Universal ACK example

7E8001000A089022849248024902480002006C7E

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5.5 GPS Data【0200】

5.5.1 GPS Data 0x0200 Structure [Table 5-5]

7E0200002C013654485438000D00000100000000003015A941F06CF58F6002600000000230619160
8400104000000000300119310116E102012A567E

Index	Message	field details	Value (HEX)	Instruction
1	Packet header		7E	It is fixed to be 0x7E
2	Message header	Message ID	0200	It is fixed to be 0x0200
3		Message body's attribute	002C	Message data length
4		Terminal ID	013654485438	Terminal ID,BCD code
5		Message serial number	000D	Message serial number
6	Message- basic location and basic information	Alarm indicator	00000100	Details with reference. refer to Table 5-6
7		Status indicator definition	00000003	Details with reference. refer to Table 5-7
8		Latitude	015A941F	Convert 015A941F to decimal data (22713375/1000000)= 22.713375 unit as degree,South/ north latitude indictor refer to Table 5-7
9		Longitude	06CF58F6	Convert 06CF8A58 to decimal data (114252022/1000000)= 114.252022 Unit as degree, east/west longitude indictor refer to Table 5-7
10		Altitude	0026	Unit as meter
11		Speed	0000	Unit as 0.1KM/H
12		Direction	0000	Due North is 0, clockwise 0-359
13		Date	230619	YYMMDD. 2023-06-19 GMT time
14		Time	160840	HHNNSS. 16:08:40 GMT time
15	Message- location extra message refer to Table 5-10	Mileage	010400000000	DWORD, 1/10KM Unit is 0.1KM
		GSM Signal	300119	0x19=25,GSM Signal is 25
		GNSS Signal	310116	Number of positioning satellites,0x16=22
		Battery voltage	E102012A	unit 1/10V,0x01 0x2A=298/10=29.8V
16	Check code		56	1 byte
17	Packet end		7E	Packet end is fixed to 0x7E

Table 5-5

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5.5.2 GPS Data User-defined Item

5.5.2.1 Alarm Indicator Define [Table 5-6]

Terminal does send the GPS data in case of alarm and there is alarm mark in the location data, the platform acknowledges the alarm by Platform Universal ACK. Alarm indicator last till the alarm condition relieve, after the alarm condition relieve, terminal send GPS data with removing the alarm indicator.

Bit	Definition	Handling instructions
0	Value is 1: Emergency Alarm (SOS)	Triggered after the alarm switch is touched, and cleared after receiving the response G106/G108/G109/EM200/G201 support
1	Value is 1: Overspeed alarm	The flag is maintained until the alarm condition is released
2	Value is 1: Fatigue driving	The flag is maintained until the alarm condition is released JT/T808-201115 / 33
3	Reserve	
4	Value is 1: GNSS module failure	The flag is maintained until the alarm condition is released
5	Value is 1: GNSS antenna not connected or cut off	The flag is maintained until the alarm condition is released
6	Value is 1: GNSS antenna short circuit	The flag is maintained until the alarm condition is released
7	Value is 1: Low power supply voltage	After the event occurs, the alarm will only be reported once Main power supply has low voltage: D100/W100/G102/G108/G109 Internal battery has low battery power: G107, W100,W500,GL600
8	Value is 1: The main power supply of the terminal is powered off	After the event occurs, the alarm will only be reported once D100/G102/G108/G109/EM200 support/EM6000 support
9	Value is 1: Acc on event	After the event occurs, the alarm will only be reported once D100/G102/G108/G109/EM200 support
10	Value is 1: Acc off event	After the event occurs, the alarm will only be reported once D100/G102/G108/G109/EM200 support
11	Value is 1: The main power supply of the terminal is powered on	After the event occurs, the alarm will only be reported once EM200,EM6000 support
12	Leave the fortified area	After the event occurs, the alarm will only be reported once G109 support
13	Reserve	
14	Value is 1: Fatigue driving warning	The flag is maintained until the alarm condition is released
15	Value is 1: Vibration alarm	After the event occurs, the alarm will only be reported once D100/G102/G108/G109 support
16	Value is 1: Flip/Fall down alarm	After the event occurs, the alarm will only be reported once D100/G102/G107/G108/G109/W600 support

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17	Value is 1: Rapid acceleration	After the event occurs, the alarm will only be reported once D100/G102/G108/G109 support
18	Value is 1: Cumulative driving overtime on the day	The flag is maintained until the alarm condition is released
19	Value is 1: Overtime parking	The flag is maintained until the alarm condition is released
20	Value is 1: Bird death	After the event occurs, the alarm will only be reported once
21	Value is 1: Birds take off	After the event occurs, the alarm will only be reported once
22	Reserve	
23	Reserve	
24	Reserve	
25	Reserve	
26	Reserve	
27	Value is 1: Illegal ignition	After the event occurs, the alarm will only be reported once D100/G102/G108/G109 support
28	Value is 1: Towing alarm	After the event occurs, the alarm will only be reported once D100/G102/G108/G109 support
29	Value is 1: Collision alarm	The flag is maintained until the alarm condition is released D100/G102/G108/G109 support
30	Harsh braking	After the event occurs, the alarm will only be reported once D100/G102/G108/G109 support
31	Harsh turn	After the event occurs, the alarm will only be reported once D100/G102/G108/G109 support

Table 5-6

5.5.2.2 Status indicator definition [Table 5-7]

Bit	Status
0	0 Acc off； 1 Acc on
1	0 is not located; 1 is located
2	0 north latitude; 1 south latitude
3	0 east longitude; 1 west longitude
4	The main power supply status: 1 is disconnected 12V/24V,0 is connected 12V/24V D100/G102/G108/G109/EM200/EM6000 support
5	0 is 4G; 1 is WIFI
6	Reserve
7	Value is 1: Low power supply voltage Main power supply has low voltage: D100/W100/G102/G108/G109 Internal battery has low battery power:

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	G107, W100,W500,GL600
8	Reserve
9	Reserve
10	0: Normal/ Immobilizer Recover; 1: fuel pump cut/Remote immobilizer
13	0: Door 1 closed; 1: Door 1 open (front door) (right front door of passenger car) G108/G108P Enable AD2 as positive digital function Send command: (addevice,1,0)
14	0: Door 2 closed; 1: Door 2 open (middle door)
15	0: Door 3 closed; 1: Door 3 open (rear door) (right rear door of passenger car)
16	0: Door 4 closed; 1: Door 4 open (driver's seat door) (left front door of passenger car)
17	1:Leave the fortified area 0:In the fortified area,return to the fortified area or clear it with the clear this function command (lockpos,1,0,0) or reset fortified area command (lockpos,1,200,10,0,0)
18	0: GPS satellites are not used for positioning; 1: GPS satellites are used for positioning.
19	0: Do not use Beidou satellites for positioning; 1: Use Beidou satellites for positioning.
20	0: GLONASS satellites are not used for positioning; 1: GLONASS satellites are used for positioning.
21	0: Do not use Galileo satellites for positioning; 1: Use Galileo satellites for positioning.
22	0: The vehicle is in a stopped state; 1: The vehicle is in a running state
23-25	0: Not positioned 1: Non differential positioning 2: Differential positioning 3: Invalid PPS 4: Differential fix positioning 5: Differential float positioning 6: Estimating
26	0: Normal data 1: Blind spot data
27-28	For PET1 & Hunting dog tracker (dog icon status) 0: Still 1: Walking 2: Running Judging by GPS speed The default value for overspeed is greater than 9KM/H and lasts for more than 5 seconds. The judgment of walking and running status is based on speed >Running at 9KM/H >=3KM/H walking
29	0: Dog not barking 1: Dog barking
30	For PET1 & Hunting dog tracker (dog pointing alert) 0:Stillness 1:Vibration
31	Reserve

Table 5-7

5.5.2.3 Message-location extra message [Table 5-8]

Field	Data Type	Description
Additional Message ID	BYTE	Support extend multi Message ID; Before we **use both 0x01 and 0x33**

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Length of Additional Message	BYTE	
Additional Message	BYTE[n]	Details

Table 5-8

Additional Message ID	Length of Additional Message	Description
0x01	4	Mileage, DWORD, Unit is 0.1KM
0x04	2	…Details, please refer to **Table 5-10**
…	…	…Details, please refer to **Table 5-10**

Table 5-9

5.5.2.4 Extra message definition [Table 5-10]

Byte Description: Byte1 to ByteN from left to right. Byte[1] Byte[2] …Byte[N]

Bit Description: Bit0 to BitN from right to left. Bit[N]…Bit[1] Bit[0]

Extra Message ID	Extra Message Length	Extra Message Description
0x01	**4**	Mileage, DWORD, 1/10KM Unit is 0.1KM (G102,G106,G108,G108P,G109)
~~**0x02**~~	~~**2**~~	~~Fuel, 2 bytes, One-way fuel analog data (G102,G106,G108,G108P)~~
0x04	**2**	Battery status, 2 bytes 1st byte: 0x00 (meaning: charger is connected) 0x01 (meaning: charger is not connected) 2nd byte: battery percentage (1-100) (G107,G109-MK)
0x2B	**4**	Fuel, 4 bytes, Two-way fuel analog data 0x0000 (Not Installed),0xFFFE (Sensor failure) (G102,G106,G108,G108P)
0x30	**1**	GSM Signal, 1 byte
0x31	**1**	GNSS Signal, 1 byte, Number of positioning satellites
0x40	**11**	**1**0001BFB2C2 **2**0001BFB2C2 **3**0001BFB2C2 BYTE[1]:Event source: **1**: Sign in **2**: Sign out **3**: After sign in, then keep **3**

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<table>
<tr><td></td><td></td><td>iButton ID: 10 Bytes, 0001BFB2C2 (The last 10 digits)
RFID Tag ID: 10 Bytes,0005748237</td></tr>
<tr><td rowspan="2">0x41</td><td rowspan="2">44</td><td>*TE,35,0027,0438,8400,0027,0438,0780,0,2805#
<table>
<tr><td>Header</td><td>Version</td><td>Smoothed Liter</td><td>Smoothed height value</td><td>Signal strength</td><td>Event</td><td>Hardware status</td><td>Realtime volume</td><td>Realtime height value</td><td>Temperature</td><td>Moving status</td><td>Number of effective signals</td><td>Dip angle</td><td>Tail</td></tr>
<tr><td>*TE</td><td>35</td><td>0027</td><td>0438</td><td>84</td><td>0</td><td>0</td><td>0027</td><td>0438</td><td>0780</td><td>0</td><td>28</td><td>05</td><td>#</td></tr>
<tr><td>①</td><td>②</td><td>③</td><td>④</td><td>⑤</td><td>⑥</td><td>⑦</td><td>⑧</td><td>⑨</td><td>⑩</td><td>⑪</td><td>⑫</td><td>⑬</td><td>⑭</td></tr>
</table>
① Header: *TE
② Version: 35
③ Smoothed Liter: 0027=2.7L, unit is 0.1L
④ Smoothed height value: 0438 = 43.8mm, UL212 unit is 0.1mm, UL212-E unit is 1mm
⑤ Signal strength: 84, The actual installation should be greater than 20
⑥ Event: 0 Normal,6 Refuel,9 Siphon
⑦ Hardware status: 0 normal,1 abnormal
⑧ Realtime volume: 0027=2.7L, unit is 0.1 liter
⑨ Realtime height value: 0438 = 43.8mm, UL212 unit is 0.1mm, UL212-E unit is 1mm
⑩ Temperature: 0780, (780-400) * 0.1 = 38.0 Celsius degree
⑪ Moving status: 0 static,1 moving
⑫ Number of effective signals: 28, The actual installation should be greater than 20
⑬ Dip angle: 05, Hexadecimal, need less than 11
⑭ Tail: #</td></tr>
<tr><td><s>0x50</s></td><td><s>2</s></td><td><s>Fuel, 2 bytes, One-way fuel analog data (G102,G106,G108,G108P)</s></td></tr>
<tr><td>0x51</td><td>8</td><td>Temperature (4 channels), 8 bytes, two bytes per channel, unit: 1/10 degree
0xFFFB (Not Installed),0xFFFE (Sensor failure)
(G102, G106, G108, G108P)</td></tr>
<tr><td>0x52</td><td>1</td><td>Forward and reverse rotation, 1 byte
0: unknown;
1: forward rotation (empty vehicle)
2: reverse rotation (heavy vehicle);
3: stop
(G102, G106, G108, G108P)</td></tr>
<tr><td>0x53</td><td>1+n*8</td><td>Base station data:
the first byte is the number of base stations, followed by n base station data; base station data: 0-1 MCC; 2 MNC; 3-4 LAC; 5-6 CELLID; 6 signal strength. 2G
(G102, G106, G107, G108, G108P, W500)</td></tr>
<tr><td>0x54</td><td>1+n*7</td><td>Wifi data:
the number of wifi in the first byte, followed by n wifi data; WIFI data: 0-5 wifiMac;6</td></tr>
</table>

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		Signal strength (G102, G106, G107, G108, G108P, W500, GL600)
0x55	**8**	Load Sensor data, 8 bytes, unit: 1/10 kg (G102, G106, G108, G108P)
0x56	**2**	Battery percentage, BYTE [1], rang is 1-10 (0x01 - 0x0A) Rreserved, BYTE [2] (G107, W500)
0x57	**8**	Byte 0: Reserved Byte 1: Bit0 vibration alarm Bit1: Pseudo base-station alarm Bytes 2-7: Reserved (AT08N)
	8	State extension, 8 bytes 0-1 bytes for alarm status 2-3 bytes for switch status 4-7 bytes Reserved: Alarm status: 0x0001 Vibration alarm 0x0002 Anti disassembly alarm (W600) 0x0004 Cover opening alarm (W600)
0x58	**8**	Humidity (4 channels), 8 bytes, two bytes per channel, unit: 1/10 degree 0xFFFB (Not Installed),0xFFFE (Sensor failure) (G102, G106, G108, G108P)
0x5D	**1+n*10**	**Base station data:** The first byte is the number of base stations, followed by n base station data; Base Station Data: 0-1 MCC; 2 MNC; 3-4 LAC; 5-8 CELLID; 9 Signal Strength. 4G
0x61	**2**	External voltage of battery products, 2-bytes, unit 1/100V (EM200)
0x66	**1**	Lock Hall sensor status Sensor1,Sensor2,Sensor3,Sensor4,Sensor5 Compose 1 byte, Sensor5 is Motor controlled Hall sensor
0x88	**4**	Working hours (hour meter), value in minutes with acc on (You can also send commands to configure using G-Sensor vibration as ACC state)
0xD8	**N**	D8 // Message ID 4C // Length 04 // Number of Slave locks 008124050002 // Master Lock MAC 00000000 // Master Lock RFID 64 // Master Lock Battery Level 000020E0 // Master Lock Status And Alarms 004024050001 // Slave Lock 1 MAC

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		00000000 // Slave Lock 1 RFID 00 // Slave Lock 1 Battery Level 00100000 // Slave Lock 1 Status And Alarms 004024050002 // Slave Lock 2 MAC 00000000 // Slave Lock 2 RFID 00 // Slave Lock 2 Battery Level 00100000 // Slave Lock 2 Status And Alarms 000000000000 // Slave Lock 3 MAC 00000000 // Slave Lock 3 RFID 00 // Slave Lock 2 Battery Level 00100000 // Slave Lock 3 Status And Alarms 000000000000 // Slave Lock 4 MAC 00000000 // Slave Lock 4 RFID 00 // Slave Lock 4 Battery Level 00100000 // Slave Lock 4 Status And Alarms **Master Lock Status: Bit31 Bit30… Bit1 Bit0** Bit0: Base Station Positioning Indicator // 0: GPS, 1: Base Station; Bit1: Enter Geo-fence alert // 0: Normal, 1: Enter Geo-fence; Bit2: Exit Geo-fence alert // 0: Normal, 1: Exit Geo-fence; Bit3: Lock rope tamper alert // 0: Normal, 1: Tamper; Bit4: Vibration alert // 0: Normal, 1: Remove; Bit5: ACK indicator // 0: Normal, 1: Need ACK; Bit6: Lock Rope status // 0: Unplugging, 1: Inserted; Bit7: Motor Lock status // 0: Unlock, 1: Locked; Bit8: Unlocking alert // 0: Normal, 1: Trigger; Bit9: Wrong password alert //0: Normal, 1: Trigger (Wrong >= 5 times); Bit10: Swiping unauthorized RFID key alert // 0: Normal, 1: Trigger; Bit11: Low battery alert // 0: Normal, 1: Trigger; Bit12: Back cover Opened alert // 0: Normal, 1: Trigger; Bit13: Back Cover status // 0: Opened, 1: Closed; Bit14: Motor fault alert // 0: Normal, 1: Trigger; Bit15 – Bit31 : Reserve // fixed at 0; **Slave Lock Status and Events: Bit31 Bit30… Bit1 Bit0** Bit0: LockRopeStatus // 0: Insert, 1: Pull out; Bit1: MotorStatus // 0: Lock, 1: Unlock; Bit2: BackCoverStatus // 0: Close, 1: Open; Bit3: ChargingStatus // 0: No Charging, 1: Charging; Bit4: MagnetStatus // 0: Normal, 1: Remove;

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		Bit5 - Bit15: Reserve //; Bit16: SLockCutOff // 0: Normal, 1: Lock rope cut alarm; Bit17: SLockOpen // 0: Normal, 1: Unlock alarm; Bit18: SUnCoverBack // 0: Normal, 1: Back cap oper alarm; Bit19: Reserve //; Bit20: STimeOut // 0: Normal, 1: Slave does not report to master for at least 3 Mins; Bit21: Reserve //; Bit22: SSealTempered // 0: Normal, 1: Tempered; Bit23: SLockStuck // 0: Normal, 1: Motor is stuck; Bit24 - Bit31: Reserve //; (GL500C, GL600C, GL800C)
0xD9	**8**	D9 08 01 01 00C44313 00 00
		Even Source 1 Byte: 1 indicates Swipe Authorized RFID key; 2 indicates Swipe Unauthorized RFID key; 3 indicates Swipe the vehicle ID blinding RFID key; 4 indicates Unlock by Password from GPRS； 5 indicates the device is automatically locked 6 indicates Unlock by Password from Bluetooth
		Unlock successful/failed 1 Byte: 1 means the RFID key or password is verified, unlocked successfully, 0 means RFID key or password is not verified, refused to unlock **Note:** If <Even Source> is 2,3,5 ,this value is fixed to 0 .
		RFID key number 4 Bytes: The ID card number when swiping the card. **Note:** If the <Event Source> is 4 or 5, the value is 00000000.
		Password True/False 1 Byte: If the <Event Source> is 4, this value indicates whether the password is correct. If password is correct, it is 1, and for other types, it is fixed to 0.
		Input Wrong Password Times 1 Byte: If the <Event Source> is 4, this value indicates the number of consecutive password input errors. For other types, it is fixed to 0.
		(GL500C, GL600C, GL800C)
0xE1	**2**	battery voltage, 2-byte, unit 1/10V G107, W500,GL600 internal battery G102,G106 ,G108,G108P,G109,G109-MK detect external car's battery voltage
0xE2	**10**	the bcd code of the SIM card information ICCID, 1 byte such as: 0x89 0x86 0x02 0xB2 0x01 0x14 0x40 0x08 0x11 0x03 The corresponding SIM card is marked: 898602B2011440081103
0xE3	**1+n*6**	Accelerometer x, y, z axis data The first byte is the number of data, followed by the acceleration sensor data. Tentative second-level acquisition, all data pushed forward from the current time
0xE4	**4**	Ambient Light Sensor, Measurements: 0.01 lux to 83 k lux (Hexadecimal, magnified 100 times,after converting to decimal, it needs to be divided by 100)

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0xE5	**2**	Atmospheric pressure sensor,300 ... 1100 hPa (equiv.to +9000...-500 m above/below sea level)
0xE7	**3**	24 types alarm status,0xE7 packet alarm status bit definition

Bit	Alarm Definition	Remark
0	1: Lock rope cutting alarm	
1	1: Emergency unlocking alarm	
2	1: Tamper alarm	
3	1: Back cover opening alarm	
4	1: Vibration alarm	
5	1: Tilt alarm	
6	1: Flip over alarm	
7	1: High temperature alarm	
8	1: Low temperature alarm	
9	1: Hat removal alarm	
10	1: Silent alarm	
11	1: Drop alarm	
12	1: Fall alarm	
13	1: Impact alarm	
14	1: Climbing high warning	
15-23	Reserve	

0xE8	**3**	24 types status,0xE8 packet switch status bit definition

Bit	Status Definition	Remark
0	1: Rotary switch open	
1	1: Motor release	
2	1: Lock lever open	
3	1: Lock rod cutting	
4	1: Lock rope cutting	
5	1: Back cover opening	
6	1: Shell opening	
7	1: Tamper	
8	1: Vibration, 0: Stationary	
9	1: Sleep mode	
10	1: Deep sleep mode	
11	1: GPS power off mode	
12	1: Device power off	
13	1: Firmware OTA in progress	
14-19	Reserve	
20-23	**Network type** **2G:** 0: GSM 2: UTRAN 3: GSM W/EGPRS	

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		3G: 4: UTRAN W/HSDPA 5: UTRAN W/HSUPA 6: UTRAN W/HSDPA and HSUPA **4G:** 7: E-UTRAN 8: UTRAN HSPA+ **5G:** 9
0xE9	**1**	1-100 battery percentage.1 byte. The same as 0xEE (W600)
0xEB	**14**	Extended sub-ID length: BYTE, 2 bytes Extended sub-ID number: BYTE, 2 bytes Extended sub-id content: BYTES At present, the extended ID only supports 0x00B2, the ID length is 10 bytes, and it refers to the ICCID information of the SIM card.
0xED	**1+7*N**	1st byte: version+number of beacons first 4 bits, current version default 0, for future expansion; Last 4 bits, beacon Total number, from 0 to 16 Beacon data starting from the second byte first version beacon data packet including MAC address and signal strength (MAC address 6 bytes, signal strength 1 byte) (W600)
0xEE	**1**	1-100 battery percentage.1 byte (G107, W500, GL600)
0xEF	**1**	Wired tracker interval battery voltage, 1-byte, unit 1/10V (G109)
0xF0	**N**	DTU transparent data transmission (maximum 128 bytes) (G108P)
0xF1	**4**	Number of unlocks (GL600)
0xF2	**2**	The GL600 lock status

Bit	Define	Description
0	0: Lock open; 1: Lock closed	
4321	0000 indicates scheduled upload; 0001 represents unlocking and uploading; 0010 indicates locking and uploading; 0011 indicates low power upload; 0100 represents mobile upload; 0101 indicates impact upload;	Indicates the trigger reason for the upload of the location information. The scheduled upload is uploaded according to the time interval set by the user, and the unlock upload is triggered by the user's unlock

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				in real time. The device should give priority to uploading data packets that are not scheduled to be uploaded.
		5	0: indicates operation; 1: indicates stop operation;	
		6	0: means discharging; 1: means charging;	
0xF4	**2**	G109-MK Extend I/O Status & Alarm		
		Bit	**Define**	**Description**
		0	**IO1 Trigger Type:** 0: Negative 1: Positive	
		1	**IO2 Trigger Type:** 0: Negative 1: Positive	
		2	reserve	
		3	reserve	
		4	reserve	
		5	reserve	
		6	**IO1 Current Status:** 0: NC 1: Triggered	
		7	**IO2 Current Status:** 0: NC 1: Triggered	
		8	reserve	
		9	**IO1 Alarm Status:** 0: No alarm 1: In alarm	
		10	**IO2 Alarm Status:** 0: No alarm 1: In alarm	
		11	reserve	
		12	reserve	
		13	reserve	
		14	reserve	
		15	reserve	
0xF8	**2**	BYTE [2], the sound is collected to adc voltage value (PET1)		
0xFA	**4**	BYTE[1], working mode: 0 normal 1 test 2 upgrade 3 shutdown BYTE[2], in shutdown mode, 0: normal shutdown 1: long-term static shutdown (PET1)		
0xFB	**1**	BYTE[1]:Wake-up source: 0x00: Unknown 0x01: Boot up 0x02: Light-sensitive(Light perception has the highest priority) 0x03: Timing 0x04: Shock 0x05: Voice control (G107,W500,GL600,PET1)		
0xFC	**15**	IMEI, 38 36 36 36 3531 30 36 37 31 39 37 37 36 37 converts to 866651067197767		
0xFD	**2**	Wired tracker internal battery voltage, 2 bytes, unit 1/10V BYTE [1], Wired tracker internal battery voltage BYTE [2], Wired tracker internal battery percentage (G109,G109-MK)		

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0xFE	N	Custom Extended Data (Extended Alarm)

Start byte	field	type of data	description
0	Extra Message ID	Byte	0xFE
1	Extra Message Length	Byte	
2	Extended Mark	Byte	0x7C Fixed value,do not process extended data without this flag
3	ID 1	Byte	
4	Length 1	Byte	
5	Data 1	N	
6	ID 2	Byte	
7	Length 2	Byte	
8	Data 2	N	
5+N	ID N	Byte	
5+N+1	Length N	Byte	
5+N+N	Data N	N	

ID	Length	Data
0x01	4 Bytes	Extended alarm information Extended alarm bit definition:

bit	Alarm Definition	Remark
0	rapid acceleration	G102/G106/G108
1	rapid deceleration	G102/G106/G108
2	sharp turn	G102/G106/G108
3	Collision	G102/G106/G108
4	Tamper	
5-31	Reserve	

Table 5-10

You can use the assistant to configure which extension bytes are uploaded by the device,
This screenshot will be modified accordingly after the extension byte commands is added.

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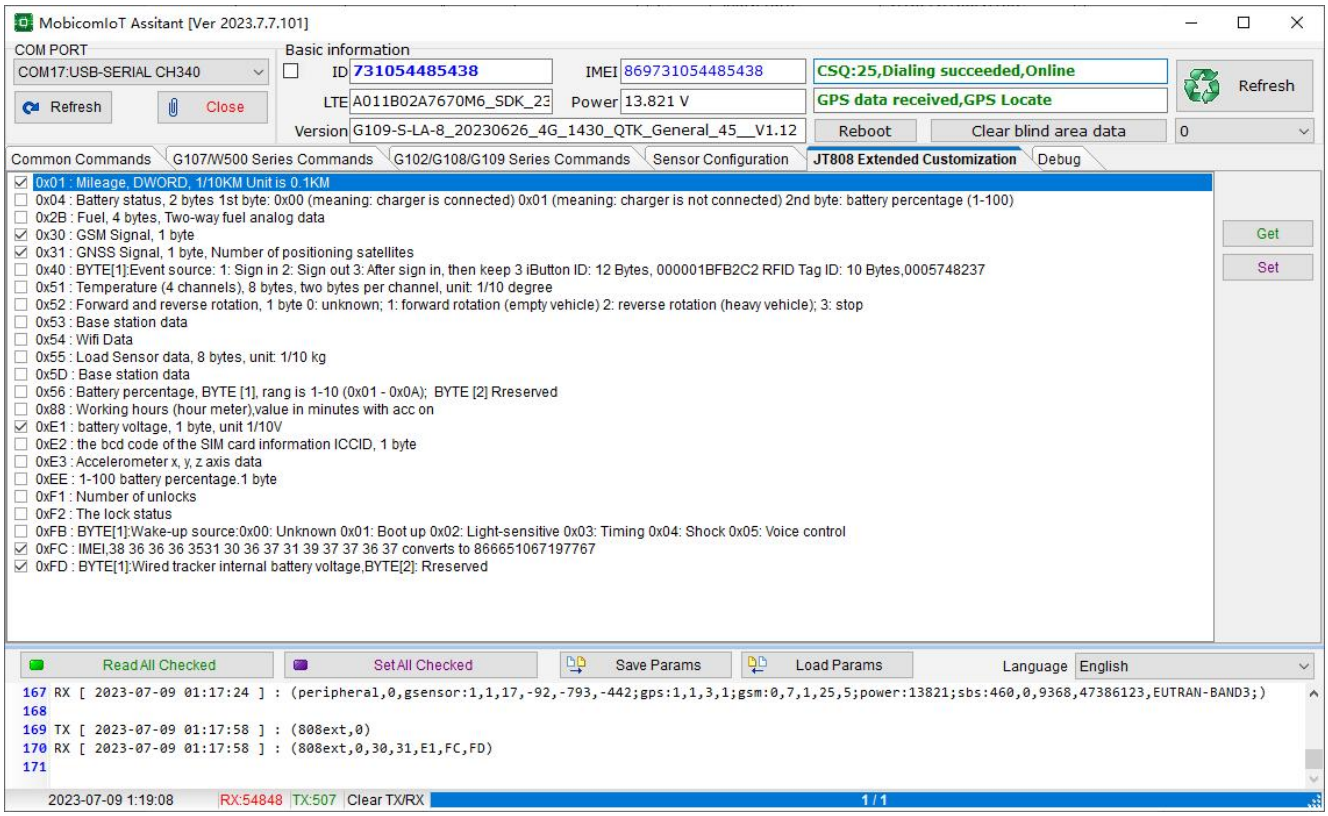

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5.6 Send Command To Tracker【8300】

Message ID:0x8300, send the command from Platform side.

Start byte	field	data type	description
0	Type	WORD	We use 0x01
1	Command body	BYTE[N]	Support max length 1024 bytes refer to **Table 5-17**

Table 5-11

5.7 Tracker Reply Command To Server【0300】

Message ID:0x0300

Start byte	field	data type	description
0	Command body	BYTE[N]	Support max length 1024 bytes

Table 5-12

Example of Command Sending and Reply:

Send To Tracker: 7E8300000A013654485438000001286375742C312C3129AC7E

Tracker Reply: 7E03000009013654485438000028637574 2C312C31292E7E

Index	Value(HEX)	Description
1	**7E**	Packet header
2	**8300**	Message ID
3	**000A**	Message body ‘s attribute, the data here means the data length is 10 bytes.
4	**013654485438**	Terminal ID
5	**0000**	Message serial number
6	**01**	transparent transmission type
7	286375742C312C3129	transparent transmission data, convert to ASCII code is (cut,1,1)
8	**AC**	Check code
9	**7E**	Protocol end

Index	Value(HEX)	Description
1	**7E**	Packet header
2	**0300**	Message ID
3	**0009**	Message body ‘s attribute, the data here means the data length is 10 bytes.
4	**013654485438**	Terminal ID
5	**0000**	Message serial number
6	286375742C312C3129	transparent transmission data, convert to ASCII code is (cut,1,1)
7	**2E**	Check code
8	**7E**	Protocol end

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5.8 Server Download Data to Tracker【8900】

Message ID:0x8900, platform send the transmission data to tracker.

Start byte	field	data type	description
0	Type	WORD	refer to **Table 5-15**
1	Command body	BYTE[N]	Support max length 1024 bytes

Table 5-13

5.9 Tracker Upload Data to Server【0900】

Message ID:0x0900, tracker send the transmission data to platform.

Start byte	field	data type	description
0	Type	WORD	refer to **Table 5-15**
1	Command body	BYTE[N]	Support max length 1024 bytes

Table 5-14

Index	Transparent message type	Value(HEX)	Description
1	GNSS Data	0x00	Detailed positioning data of GNSS module
2	IC Card Information	0x0B	The upload message of the IC card information of the road transportation certificate is 64Byte, and the download message is 24Byte. The timeout for the transparent transmission of the road transportation certificate IC card authentication is 30 seconds. After timeout, do not resend.
3	COM1	0x41	Serial port 1 transparent message
4	COM2	0x42	Serial port 2 transparent message
5	BMS1	0xB2	
6	Driving Report Data	0xF8	Driving report data transmission
7	BMS2	0xF9	
8	User defined	0xFA-0xFF	

Table 5-15

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0x8900 Example 1：

7E8900001208902284924800013332A4D30322C30382C3132333435363738230B7E

Index	Value(HEX)	Description
1	**7E**	Packet header
2	**8900**	Message ID
3	**0012**	Message body ‘s attribute, the data here means the data length is 12 bytes.
4	**019912345678**	Terminal ID
5	**0001**	Message serial number
6	**41**	transparent transmission type 0x41 mean’s COM1 0x42 mean’s COM2
7	**48656C6C6F205** **76F726C6421**	transparent transmission data, convert to ASCII code is **Hello World!**
8	**0B**	Check code
9	**7E**	Protocol end

Table 5-16

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5.10 Raw Data Example

Red: Tracker Upload Data to Server
Green: Server Send Data or Reply Data to Tracker

7E**0102**000A01365448543800044D6F6269636F6D496F54707E --- Tracker upload authorization data

7E**8001**0005013654485438000500040102 00C17E --- Server ACK

7E**0200**002C013654485438000D00000100000000003015A941F06CF58F6002600000000023061916084001040
00000003001193101 16E102012A567E --- tracker upload GPS data

7E**8001**0005013654485438000900080200 00C07E --- Server ACK

7E**8300**000A013654485438000001286375742C312C3129AC7E --- Server send command to tracker

7E**0300**00090136544854380000286375742C312C31292E7E --- Tracker reply command to server

7E**8001**0005013654485438000100000300 00C17E --- Server ACK

7E --- Packet header
0200 --- Message ID
002C --- Message data length
013654485438 --- Terminal ID,BCD code
000D --- Message serial number
00000100 --- Alarm indicator
00000003 --- Status indicator definition
015A941F --- Latitude
06CF58F6 --- Longitude
0026 --- Altitude Unit as meter
0000 --- Speed Unit as 0.1KM/H
0000 --- Direction Due North is 0, clockwise 0-359
230619 --- Date YYMMDD
160840 --- Time HHNNSS
01 04 00000000 --- Mileage, DWORD, 1/10KM Unit is 0.1KM
30 01 19 --- GSM Signal, 1 byte
31 01 16 --- GNSS Signal, 1 byte, Number of positioning satellites
E1 02 012A --- battery voltage, 2-byte, unit 1/10V
56 --- Check code
7E --- Packet end

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7E --- Packet header
8300 --- Message ID
000A --- Message data length
013654485438 -- Terminal ID,BCD code
0000 --- Message serial number
01 --- Type fixed 0x01
286375742C312C3129 --- command body, convert hexadecimal to ASCII is (cut,1,1)
AC --- Check code
7E --- Packet end

7E --- Packet header
0300 --- Message ID
0009 --- Message data length
013654485438 -- Terminal ID,BCD code
0000 --- Message serial number
286375742C312C3129 --- command body, convert hexadecimal to ASCII is (cut,1,1)
2E --- Check code
7E --- Packet end

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5.11 Common Commands [Table 5-17]

For more commands, please refer to Mobicom Telematics Command List Excel file.

Command & Description Format is ' (commandKey,0/1,param1,param2,paramN) ',word case doesn't matter.
Command contains left '(' and right ')' parentheses!!!

1. Query/Set Vibration Mode:

Vibration wakes up default value is 80mg (can be modified), after wake up the max working time is 180 seconds

Mode A: Vibration upload interval, minimum 5S, maximum 180S, turning point upload, GPS on
Mode B: No vibration upload interval, minimum 600S, maximum 86400S (24 hours), GPS off

Query: (move,0)
Reply: (move,0,5,600) vibration upload interval 5 seconds, no vibration upload interval 600 seconds
Set: (move,1,5,600) vibration upload interval 5 seconds, no vibration upload interval 600 seconds
Reply: (move,1,5,600)

2. Query/Set Timing Mode:

GPS is turned on by default, A & B mode can be superimposed

Mode A: Fixed time point mode, 0-24 hours (0,1,2,3....24)
Mode B: Period grouping mode, start time, end time, upload times

Mode A set command: (time,1,a,0,1,2,3,4,5,6,7,8,9,10,11,12,13...23) - fixed time point mode 0-23 hours
Reply: (time,1,a,0,1,2,3,4....23)
Mode A clear command: (time,1,a) - fixed time point mode clear
Reply: (time,1,a)

Mode B time period set 1: (time,1,b,1,0800,0900,4) --start time 8:00, end time 9:00, upload times 4 times
Reply: (time,1,b,1,0800,0900,4)

Mode B time period set 2: (time,1,b,2,1300,1400,5) --start time 13:00, end time 14:00, upload times 5 times
Reply: (time,1,b,2,1300,1400,5)

Clear all time points in B mode: (time,1,b) --Clear all time points in time period grouping mode
Reply: (time,1,b)

Query: (time,0)
Reply: (time,0,a,0,1,2,3,4....23;b1,0800,0900,4;b2,1300,1400,5)

3. Query/Set time zone

Set:(timezone,1,8:00)

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Reply: (timezone,1,8:00)

Set:(timezone,1,-9:30)
Reply: (timezone,1,-9:30)

Query: (timezone,0)
Reply: (timezone,0,-9:30)

4. Query/Set G-Sensor detects vibration judgment conditions

Can adjust the G-Sensor judgment conditions for detecting vibration wake-up

Set:(gsensor,1,123,10)
Reply: (gsensor,1,123,10)

The first parameter is the mg value, and the second parameter is the number of consecutive occurrences greater than this mg value.

Query: (gsensor,0)
Reply: (gsensor,0,123,10)

5. Query/Set APN

Set: (apn,1,cmnet,user,password)
Reply: (apn,1,cmnet,user,password)

Query: (apn,0)
Reply: (apn,0,cmnet,user,password)

6. Query/Set Main/Backup IP

1 mean's main IP/Domain, 2 mean's backup

Set IP1: (ip,1,1,168.192.18.18,8003)
Reply: (ip,1,1,168.192.18.18,8003)

Set Domain name 2: (ip,1,2,xxxx.com,8004)
Reply: (ip,1,2,xxxx.com,8004)

Query: (ip,0)
Reply: (ip,0,1,168.192.18.18,8003;2,xxxx.com,8004)

7. Query/Set Tracker's ID

Set ID: (id,1,019912345678)
Reply: (id,1,019912345678)

Query ID: (id,0)
Reply:(id,0,019912345678)

8. Query general parameters

Reply format: (all,0,*commandKey1,params1*commandKey2,params2....)

E.g. IP port,ID,APN,TCP/UDP,ACC upload interval,backup battery,main battery,online status,time zone,GSM signal,GPS signal, number of satellites, software version....

Query: (all,0) or (all)

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Reply: (all,0,*move,0,5,600*timezone,0,9:30*apn,0,cmnet,user,password*ip,0,1,168.192.18.18,8003;2,xxxx.com,8004*net ,0,tcp*id,0,019912345678)
9. Reboot **Reset:** (reboot,1) **Reply:** (reboot,1)
10. Factory reset **Factory reset:** (reset,1) **Reply:** (reset,1)
11. Remote upgrade OTA **Remote upgrade:** (OTA,1,148.70.5.127,8122) **Reply:** (OTA,1,148.70.5.127,8122)
12. version query **Version query:** (ver,0) **Reply:** (ver,0,version number)
13. JT808 protocol extended additional information customization, in 0x0200 upload data **Set :** (808ext,1,01,04,30,31,56,E1,FE) **Reply:** (808ext,1,01,04,30,31,E1,EF,F6,FB) **Query:** (808ext,0) **Reply:** (808ext,0,30,31,E1,EF,F6,FB)

Table 5-17

5.12 The second configuration parameter method 0x8103

Message ID: 0x8103, set the parameter message body data format as shown:

Start byte	field	data type	description
0	total number of parameters	WORD	
1	parameter list	parameter format refers to Table 5-19	

Table 5-18 Parameter message body data format

Field	Data type	Description
parameter ID	DWord	Parameter ID definition and description refers to Table 5-20
Parameter length	Byte	
Parameter value		If it is a multi-value parameter, use multiple parameter items with the same ID in the message

Table 5-19 parameter format

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Parameter ID	Data type	Description
0x0001	DWORD	Tracker heartbeat sending interval, in seconds (s)
0x0002	DWORD	TCP message response timeout, in seconds (s)
0x0003	DWORD	TCP message retransmission times
0x0010	STRING	APN
0x0011	STRING	APN username
0x0012	STRING	APN password
0x0013	STRING	Server IP or domain
0x0018	DWORD	Server TCP port
0x0029	DWORD	Reporting upload interval, in seconds (s), >0
0x0055	DWORD	Maximum speed (km/h)
0x0056	DWORD	Overspeed duration, in seconds (s)
0x0080	DWORD	Vehicle odometer reading, 1/10km

Table 5-20 Parameter ID definition and description

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